

1 Introduction/Background

Definition 1.1. A closed oriented n -manifold M is called enlargeable if there exists a Riemannian metric on M , such that for all $\varepsilon > 0$ there exists a finite covering $\hat{M} \rightarrow M$ together with a non-zero degree ε -contracting map $\hat{M} \rightarrow S^n$.

Note that this notion does not depend on the Riemannian metric on M . Our definition of enlargeability is adopted from [?] in contrast to the original definition by Gromov and Lawson in [?], where the covering $\hat{M}$ was additionally required to be spin.

The following Lemma provides two easy ways to construct new enlargeable manifolds out of given ones.

Lemma 1.2. *i) The product of enlargeable manifolds is enlargeable.*

ii) A closed oriented manifold M admitting a non-zero degree map onto an enlargeable manifold is enlargeable.

Theorem 1.3 (Gromov, Lawson and Cecchini, Schick). *Enlargeable manifolds do not admit metrics of positive scalar curvature.*

Question 1.4. Do there exist circle bundles $E \rightarrow M$ over enlargeable base spaces M , such that the total space E admits a metric of positive scalar curvature?

In the following we will give a positive answer to the above question by constructing examples in all dimensions ≥ 4 . The tool for constructing a positive scalar curvature metric on E will be provided by the following theorem.

Theorem 1.5 (Gromov, Hanke). ...

2 Preliminaries

- Chern and Stiefel-Whitney classes
- 1:1 correspondence of $\mathbb{K}$ vector bundles and $GL(n, \mathbb{K})$ principal bundles + in particular complex line bundles and S^1 -principal bundles ($\rightarrow$ two different constructions either as underlying principal bundle or by choosing metric) (
- example: underlying principal bundle of canonical line bundle over $\mathbb{C}P^n$ is $S^{2n+1} \rightarrow \mathbb{C}P^n$)
- 1:1 correspondence of S^1 -principal bundles over M and second cohomology classes of M
- Definition of spin/non-spin/totally-non-spin (Example: $\mathbb{C}P^2$ is not spin)
- Lemma: If M contains a codimension 0 (immersed) submanifold which is totally non-spin, then M is itself totally non-spin
- Definieren die classifying maps $X \rightarrow B\pi_1(X)$ eine natürliche Transformation? (Für Kommutativität von dem einen Diagram)

3 The Construction

Let N^{2n} be any $2n$ -dimensional totally non-spin manifold, $n \geq 2$. We could for example take $N := \mathbb{C}P^2 \times T^{2n-4}$. Consider the manifold

$$M := \mathbb{C}P^n \# T^{2n} \# N.$$

Since the collapse map $M \rightarrow T^{2n}$ has degree 1 and T^{2n} is enlargeable, it follows from Lemma 1.2 that M is also enlargeable. We will now construct a circle bundle $E \rightarrow M$ as follows. Let

$$c_1 \in H^2(M; \mathbb{Z}) \cong H^2(\mathbb{C}P^n; \mathbb{Z}) \oplus H^2(T^{2n}; \mathbb{Z}) \oplus H^2(N; \mathbb{Z})$$

be the cohomology class corresponding to the first chern class $\alpha \in H^2(\mathbb{C}P^n; \mathbb{Z})$ of the canonical bundle $S^{2n+1} \rightarrow \mathbb{C}P^n$. We then define $E \rightarrow M$ as the unique S^1 -principal bundle with first chern class equal to c_1 . Now this is quite an abstract definition, so let us develop an intuition for how this bundle looks like. We claim that E is trivial over the summand $T^{2n} \# N$ and is given by the restriction of the canonical bundle $S^{2n+1} \rightarrow \mathbb{C}P^n$ over the $\mathbb{C}P^n$ summand. To make this precise let

$$i_1: (T^{2n} \# N) - D^{2n} \rightarrow M$$

be the inclusion. Then

$$\begin{array}{ccc} H^2(M; \mathbb{Z}) & \xrightarrow{i_1^*} & H^2((T^{2n} \# N) - D^{2n}; \mathbb{Z}) \\ \cong \uparrow & & \cong \uparrow \\ H^2(\mathbb{C}P^n; \mathbb{Z}) \oplus H^2(T^{2n} \# N; \mathbb{Z}) & \xrightarrow{\text{proj.}} & H^2(T^{2n} \# N; \mathbb{Z}) \end{array}$$

commutes and therefore $i_1^*(c_1) = 0$. Again using the fact that S^1 -principal bundles are uniquely determined by their first chern class we see that i_1^*E is indeed trivial. Similiarly if we denote by

$$i_2: \mathbb{C}P^n - D^{2n} \rightarrow M, \quad j: \mathbb{C}P^{2n} - D^{2n} \rightarrow \mathbb{C}P^{2n}$$

the inclusions, the diagram

$$\begin{array}{ccc} H^2(M; \mathbb{Z}) & \xrightarrow{i_2^*} & H^2(\mathbb{C}P^n - D^{2n}; \mathbb{Z}) \\ \cong \uparrow & & \cong \uparrow j^* \\ H^2(\mathbb{C}P^n; \mathbb{Z}) \oplus H^2(T^{2n} \# N; \mathbb{Z}) & \xrightarrow{\text{proj.}} & H^2(\mathbb{C}P^n; \mathbb{Z}) \end{array}$$

commutes and thus $i_2^*(c_1) = j^*(\alpha)$. So the first chern classes of i_2^*E and j^*S^{2n+1} agree, hence they must be isomorphic bundles.

With this intuition in mind we now want to show the existence of a metric of positive scalar curvature on E by applying Theorem 1.5. In order to apply the Theorem we will show the following two claims.

Claim 1. E is totally non-spin.

Claim 2. The projection $p: E \rightarrow M$ induces an isomorphism on π_1 .

Granting these Claims for now let us see why they suffice for the assumptions of Theorem 1.5 to be met. For this we would like to show that E is null-homologous as a cycle in the group homology $H_{2n}(\pi_1(E); \mathbb{Z})$, which can be seen by using Claim 2 as follows. Let $q: DE \rightarrow M$ be the disk bundle corresponding to E . Moreover let $\phi: E \rightarrow B\pi_1(E)$ and $\Phi: DE \rightarrow B\pi_1(DE)$ be the classifying maps of the respective universal coverings. Then the diagram

$$\begin{array}{ccc} DE & \xrightarrow{\Phi} & B\pi_1(DE) \\ \uparrow \iota & & \uparrow B\pi_1(\iota) \\ E & \xrightarrow{\phi} & B\pi_1(E) \end{array}$$

commutes (Begründung rein!), where the vertical maps are (induced by) the inclusion $\iota: E \rightarrow DE$. We claim that $B\pi_1(\iota)$ is an isomorphism. This follows from commutativity of

$$\begin{array}{ccc} B\pi_1(E) & \xrightarrow{B\pi_1(\iota)} & B\pi_1(DE) \\ & \searrow \cong & \swarrow \cong \\ & B\pi_1(p) & B\pi_1(q) \\ & \searrow & \swarrow \\ & B\pi_1(M) & \end{array}$$

where $B\pi_1(q)$ is an isomorphism by homotopy invariance and $B\pi_1(p)$ is an isomorphism by Claim 2. Thus ϕ factors over ι . Since DE is an oriented null-cobordism of E it follows that $\iota_*[E] = 0 \in H_{2n}(DE; \mathbb{Z})$ and so in particular $\phi_*[E] = 0 \in H_{2n}(\pi_1(E); \mathbb{Z})$ as we wanted to show. So together with Claim 1 all assumptions of Theorem 1.5 are met, showing that E admits a metric of positive scalar curvature. We will now proof Claim 1 and Claim 2

Proof of Claim 1. Let $i: N - D^{2n} \rightarrow M$ be the inclusion. We have already established above, that E is trivial over the $T^{2n} \# N$ summand and so in particular

$$i^*E \cong (N - D^{2n}) \times S^1.$$

Consequently the universal cover of i^*E is given by $\tilde{N} \times \mathbb{R}$, where $\tilde{N}$ is the universal cover of N . Let $\pi_{\tilde{N}}: \tilde{N} \times \mathbb{R} \rightarrow \tilde{N}$ and $\pi_{\mathbb{R}}: \tilde{N} \times \mathbb{R} \rightarrow \mathbb{R}$ be the projections. We compute

$$w_2(T(\tilde{N} \times \mathbb{R})) = w_2(\pi_{\tilde{N}}^* T\tilde{N} \oplus \pi_{\mathbb{R}}^* T\mathbb{R}) = \pi_{\tilde{N}}^* w_2(T\tilde{N}).$$

Since $\pi_{\tilde{N}}$ is a homotopy equivalence and $\tilde{N}$ is by construction non-spin, it follows that $w_2(T(\tilde{N} \times \mathbb{R})) \neq 0$, such that i^*E is totally non-spin. Therefore M contains a totally non-spin submanifold of codimension 0 and is consequently itself totally non-spin by (Lemma in Preliminaries section einfügen) $\square$

Proof of Claim 2. Let $j: \mathbb{C}P^n - D^{2n} \rightarrow M$ be the inclusion. Consider the following commutative diagram, where the rows are taken from the long exact sequences of homotopy groups of the respective fibrations and the vertical maps are induced by inclusions:

$$\begin{array}{ccccc} \pi_2(M) & \xrightarrow{\partial_1} & \pi_1(S^1) & \longrightarrow & \pi_1(E) \\ \uparrow & & \text{id} \uparrow & & \uparrow \\ \pi_2(\mathbb{C}P^n - D^{2n}) & \xrightarrow{\partial_2} & \pi_1(S^1) & \longrightarrow & \pi_1(j^*E) \\ \cong \downarrow & & \text{id} \downarrow & & \downarrow \\ \pi_2(\mathbb{C}P^n) & \xrightarrow{\partial_3} & \pi_1(S^1) & \longrightarrow & \pi_1(S^{2n+1}) \end{array}$$

The vertical map on the lower left is an isomorphism, because both $\mathbb{C}P^n$ and $\mathbb{C}P^n - D^{2n}$ are simply connected and so by the Hurewicz Theorem

$$\begin{array}{ccc} \pi_2(\mathbb{C}P^n - D^{2n}) & \longrightarrow & \pi_2(\mathbb{C}P^n) \\ \downarrow \cong & & \downarrow \cong \\ H_2(\mathbb{C}P^n - D^{2n}; \mathbb{Z}) & \xrightarrow{\cong} & H_2(\mathbb{C}P^n; \mathbb{Z}) \end{array}$$

commutes and the vertical maps are isomorphisms. We now claim that ∂_1 is surjective. For this it suffices to show surjectivity of ∂_2 , which in turn is equivalent to surjectivity of ∂_3 . But this is clear by exactness of the rows and $\pi_1(S^{2n+1}) = 0$. Consequently

$$\pi_1(S^1) \rightarrow \pi_1(E)$$

is the zero map and therefore we conclude from exactness of

$$\pi_1(S^1) \rightarrow \pi_1(E) \rightarrow \pi_1(M) \rightarrow \pi_0(S^1),$$

that $\pi_1(E) \rightarrow \pi_1(M)$ is indeed an isomorphism. $\square$

The above construction gives examples to Question 1.4 in all even dimensions ≥ 4 . To also get examples in odd dimensions let $E \rightarrow M$ be as above and let $\pi: M \times S^1 \rightarrow M$ be the projection. Then the pullback bundle $\pi^*E \cong E \times S^1$ clearly admits a metric of positive scalar curvature and by Lemma 1.2 the base space $M \times S^1$ is still enlargeable.